THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 10: **How Quantities Combine (Understanding Data)** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 42	Sum 210 divided by 5.
2	**(c)** 40	Equal-sized groups: average of the averages.
3	**(b)** 87	29% of $300 = \frac{29}{100} \times 300 = 87$.
4	**(a)** 95	Total length is the sum of the components.
5	**(b)** 51	$\frac{24 \times 46 + 15 \times 59}{39} = 51$.
6	**(c)** 30 mL	$\frac{12}{100} \times 250 = 30$.
7	**(c)** total of all values divided by the total number of values	Use totals, not the plain mean of the two averages.
8	**(b)** 72	$\frac{79 \times 2 + 64 \times 1 + 69 \times 2}{5} = 72$.
9	**(d)** 33	Sum 198 divided by 6.
10	**(c)** 43	Equal-sized groups: average of the averages.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Sum $= 55 + 27 + 50 + 44 + 54 = 230$ (1); Mean $= \frac{230}{5} = 46$ (1).	1+1
12	Food $= \frac{25}{100} \times 200 = 50$; Remaining share $= 75\%$ (1); Everything else $= 200 - 50 = 150$ (1).	1+1
13	Pulp $= \frac{36}{100} \times 250 = 90$ mL (1); Remaining $= 250 - 90 = 160$ mL (1).	1+1
14	Sum $= 44 + 36 + 42 + 30 = 152$ (1); Mean $= \frac{152}{4} = 38$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Total of first group $= 10 \times 47 = 470$ (1); Total of second group $= 30 \times 31 = 930$ (1); Combined average $= \frac{1400}{40} = 35$ (1).	1+1+1
16	Sum of weekly averages $= 123$ (1); Average $= \frac{123}{3} = 41$ min per day (1); Every week has the same number of days, so each average has the same weight (1).	1+1+1
17	Weighted sum $= 83 \times 3 + 61 \times 3 = 432$ (1); Sum of weights $= 6$ (1); Weighted average $= 72$ (1).	1+1+1
	OR: Total of first group $= 24 \times 46 = 1104$ (1); Total of second group $= 15 \times 59 = 885$ (1); Combined average $= \frac{1989}{39} = 51$ (1).	1+1+1
18	Sum of weekly averages $= 225$ (1); Average $= \frac{225}{5} = 45$ min per day (1); Every week has the same number of days, so each average has the same weight (1).	1+1+1
	OR: Weighted sum $= 42 \times 2 + 58 \times 2 = 200$ (1); Sum of weights $= 4$ (1); Weighted average $= 50$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Sum of 12 observations $= 12 \times 50 = 600$ (1); New sum $= 600 + 63 = 663$ (2); New mean $= \frac{663}{13} = 51$ (2).	1+2+2
	OR: Pure part in the first $= 20 \times \frac{40}{100} = 8$ mL (1); Pure part in the second $= 16.8$ mL (1); Total $= 24.8$ mL in 80 mL (1); Concentration $= 31\%$ (2).	1+1+1+2
20	Sum of 10 observations $= 10 \times 35 = 350$ (1); New sum $= 350 + 57 = 407$ (2); New mean $= \frac{407}{11} = 37$ (2).	1+2+2
	OR: Pure part in the first $= 60 \times \frac{28}{100} = 16.8$ mL (1); Pure part in the second $= 4.8$ mL (1); Total $= 21.6$ mL in 80 mL (1); Concentration $= 27\%$ (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 10: **How Quantities Combine (Understanding Data)** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 32	32% of $100 = \frac{32}{100} \times 100 = 32$.
2	**(d)** 98	Total length is the sum of the components.
3	**(a)** 42	$\frac{24 \times 41 + 12 \times 44}{36} = 42$.
4	**(b)** 20 mL	$\frac{5}{100} \times 400 = 20$.
5	**(c)** the two groups have the same size	Equal weights.
6	**(a)** 67	$\frac{85 \times 2 + 51 \times 3 + 79 \times 1}{6} = 67$.
7	**(c)** 24	Sum 120 divided by 5.
8	**(c)** 40	Equal-sized groups: average of the averages.
9	**(d)** 35	25% of $140 = \frac{25}{100} \times 140 = 35$.
10	**(c)** 84	Total length is the sum of the components.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Food $= \frac{30}{100} \times 140 = 42$; Remaining share $= 70\%$ (1); Everything else $= 140 - 42 = 98$ (1).	1+1
12	Pulp $= \frac{28}{100} \times 350 = 98$ mL (1); Remaining $= 350 - 98 = 252$ mL (1).	1+1
13	Sum $= 12 + 17 + 62 + 42 + 68 + 15 = 216$ (1); Mean $= \frac{216}{6} = 36$ (1).	1+1
14	Food $= \frac{15}{100} \times 160 = 24$; Remaining share $= 85\%$ (1); Everything else $= 160 - 24 = 136$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Total of first group $= 26 \times 39 = 1014$ (1); Total of second group $= 18 \times 61 = 1098$ (1); Combined average $= \frac{2112}{44} = 48$ (1).	1+1+1
16	Sum of weekly averages $= 208$ (1); Average $= \frac{208}{4} = 52$ min per day (1); Every week has the same number of days, so each average has the same weight (1).	1+1+1
17	Weighted sum $= 51 \times 2 + 95 \times 2 = 292$ (1); Sum of weights $= 4$ (1); Weighted average $= 73$ (1).	1+1+1
	OR: Total of first group $= 26 \times 61 = 1586$ (1); Total of second group $= 13 \times 58 = 754$ (1); Combined average $= \frac{2340}{39} = 60$ (1).	1+1+1
18	Sum of weekly averages $= 144$ (1); Average $= \frac{144}{3} = 48$ min per day (1); Every week has the same number of days, so each average has the same weight (1).	1+1+1
	OR: Weighted sum $= 95 \times 2 + 89 \times 2 = 368$ (1); Sum of weights $= 4$ (1); Weighted average $= 92$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Sum of 7 observations $= 7 \times 41 = 287$ (1); New sum $= 287 + 57 = 344$ (2); New mean $= \frac{344}{8} = 43$ (2).	1+2+2
	OR: Pure part in the first $= 40 \times \frac{18}{100} = 7.2$ mL (1); Pure part in the second $= 4.8$ mL (1); Total $= 12$ mL in 60 mL (1); Concentration $= 20\%$ (2).	1+1+1+2
20	Sum of 9 observations $= 9 \times 44 = 396$ (1); New sum $= 396 + 54 = 450$ (2); New mean $= \frac{450}{10} = 45$ (2).	1+2+2
	OR: Pure part in the first $= 40 \times \frac{27}{100} = 10.8$ mL (1); Pure part in the second $= 13.2$ mL (1); Total $= 24$ mL in 100 mL (1); Concentration $= 24\%$ (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 10: **How Quantities Combine (Understanding Data)** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 48	$\frac{24\times42+18\times56}{42}=48.$
2	**(b)** 90 mL	$\frac{15}{100}\times600=90.$
3	**(d)** the total of all the components	Whole bar = total.
4	**(d)** 79	$\frac{72\times3+76\times2+88\times3}{8}=79.$
5	**(c)** 37	Sum 148 divided by 4.
6	**(a)** 30	Equal-sized groups: average of the averages.
7	**(c)** 75	25% of $300=\frac{25}{100}\times300=75.$
8	**(a)** 80	Total length is the sum of the components.
9	**(b)** 55	$\frac{28\times59+14\times47}{42}=55.$
10	**(a)** 280 g	$\frac{35}{100}\times800=280.$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Pulp $=\frac{19}{100}\times300=57$ mL (1); Remaining $=300-57=243$ mL (1).	1+1
12	Sum $=70+21+69+44=204$ (1); Mean $=\frac{204}{4}=51$ (1).	1+1
13	Food $=\frac{35}{100}\times160=56$; Remaining share $=65\%$ (1); Everything else $=160-56=104$ (1).	1+1
14	Pulp $=\frac{24}{100}\times350=84$ mL (1); Remaining $=350-84=266$ mL (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Total of first group $=18\times64=1152$ (1); Total of second group $=12\times49=588$ (1); Combined average $=\frac{1740}{30}=58$ (1).	1+1+1
16	Sum of weekly averages $=135$ (1); Average $=\frac{135}{3}=45$ min per day (1); Every week has the same number of days, so each average has the same weight (1).	1+1+1
17	Weighted sum $=40\times2+85\times4=420$ (1); Sum of weights $=6$ (1); Weighted average $=70$ (1).	1+1+1
	OR: Total of first group $=25\times30=750$ (1); Total of second group $=20\times48=960$ (1); Combined average $=\frac{1710}{45}=38$ (1).	1+1+1
18	Sum of weekly averages $=132$ (1); Average $=\frac{132}{4}=33$ min per day (1); Every week has the same number of days, so each average has the same weight (1).	1+1+1
	OR: Weighted sum $=57\times4+91\times4=592$ (1); Sum of weights $=8$ (1); Weighted average $=74$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Sum of 5 observations $=5\times29=145$ (1); New sum $=145+59=204$ (2); New mean $=\frac{204}{6}=34$ (2).	1+2+2
	OR: Pure part in the first $=30\times\frac{22}{100}=6.6$ mL (1); Pure part in the second $=11.4$ mL (1); Total $=18$ mL in 90 mL (1); Concentration $=20\%$ (2).	1+1+1+2
20	Sum of 6 observations $=6\times47=282$ (1); New sum $=282+61=343$ (2); New mean $=\frac{343}{7}=49$ (2).	1+2+2
	OR: Pure part in the first $=20\times\frac{30}{100}=6$ mL (1); Pure part in the second $=9.6$ mL (1); Total $=15.6$ mL in 60 mL (1); Concentration $=26\%$ (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.